

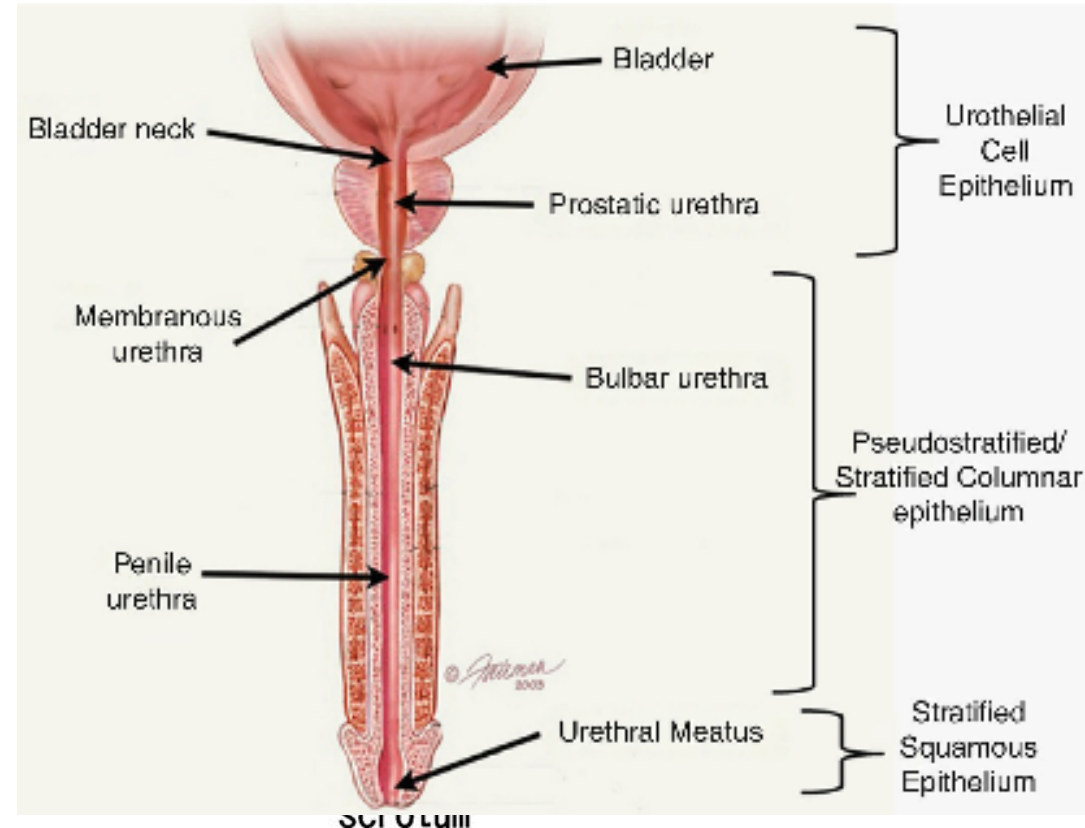
ANATOMY OF URETHRA

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MALE URETHRA

- The male urethra is a muscular tube (18–22 cm long).
- Conveys urine from the internal urethral orifice to external urethral orifice.
- Provides an exit for semen (sperms and glandular secretions).
- It divided into four parts:
 - i. Intramural (pre prostatic).
 - ii. Prostatic.
 - iii. Intermediate (membranous).
 - iv. Spongy (penile).



MALE URETHRA

TABLE 3.6. PARTS OF MALE URETHRA

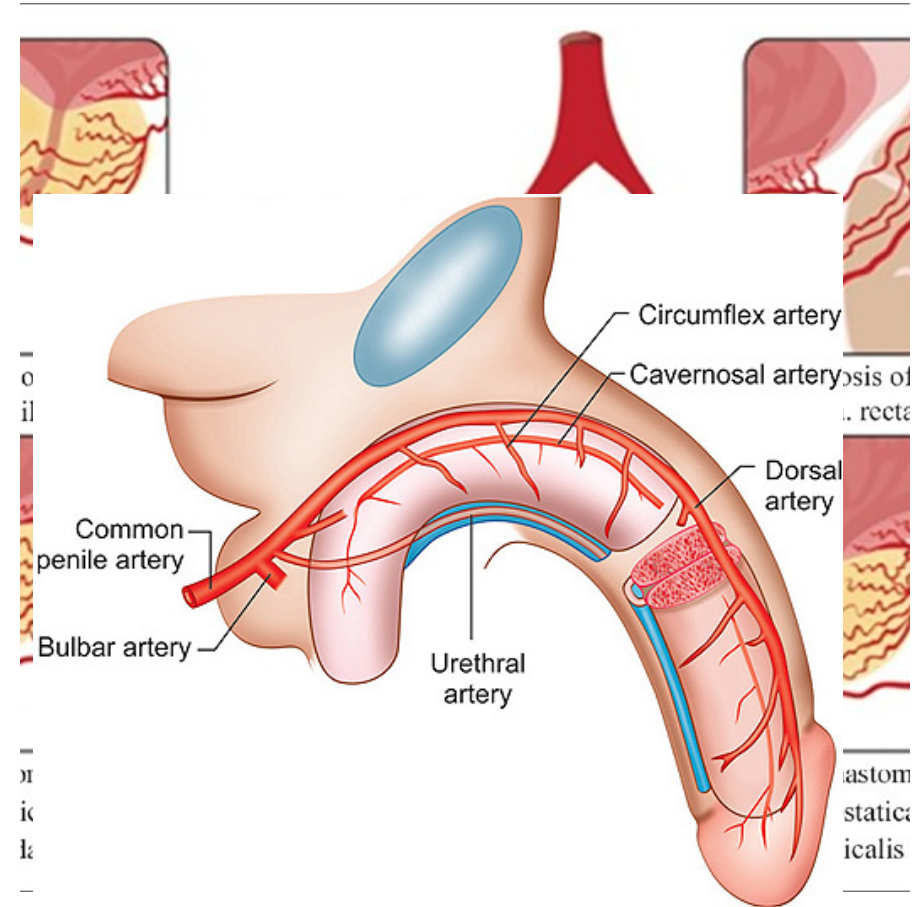
Part	Length ^a	Location/Disposition	Features
Intramural (preprostatic) part	0.5-1.5 cm	Extends almost vertically through neck of bladder	Surrounded by internal urethral sphincter; diameter and length vary, depending on whether bladder is filling or emptying
Prostatic urethra	3.0-4.0 cm	Descends through anterior prostate, forming a gentle, anteriorly concave curve; is bounded anteriorly by a vertical trough-like part (rhabdosphincter) of external urethral sphincter	Widest and most dilatable part; features urethral crest with seminal colliculus, flanked by prostatic sinuses into which prostatic ducts open; ejaculatory ducts open onto colliculus, hence urinary and reproductive tracts merge in this part
Intermediate (membranous) part	1.0-1.5 cm	Passes through deep perineal pouch, surrounded by circular fibers of external urethral sphincter; penetrates perineal membrane	Narrowest and least distensible part (except for external urethral orifice)
Spongy urethra	~ 15 cm	Courses through corpus spongiosum; initial widening occurs in bulb of penis; widens again distally as navicular fossa (in glans penis)	Longest and most mobile part; bulbourethral glands open into bulbous part; distally, urethral glands open into small urethral lacunae entering lumen of this part



MALE URETHRA

❑ Blood Supply:

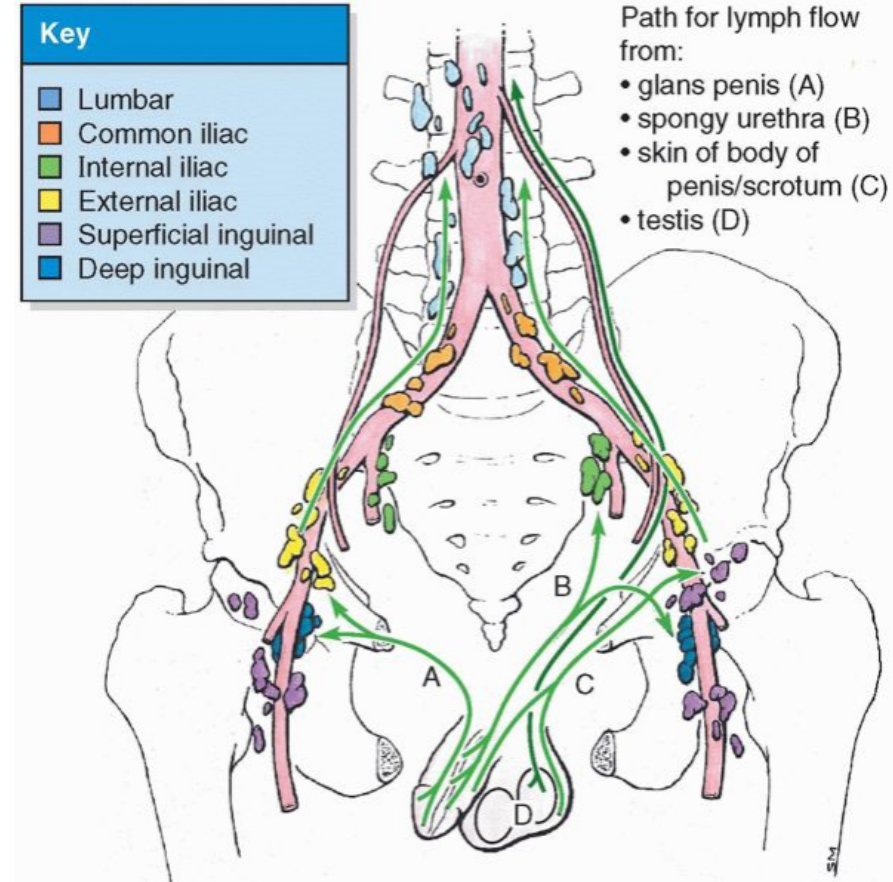
- Intramural & prostatic parts are supplied by:
- ✓ Prostatic vessels {from inferior vesical & middle rectal vessels}.
- Blood supply of intermediate and spongy parts supplied by:
- ✓ Dorsal vessels of the penis.



MALE URETHRA

□ Lymphatics:

- Lymph from urethra pass to:
- ✓ Internal iliac lymph nodes.
- ✓ Deep inguinal lymph nodes.



MALE URETHRA

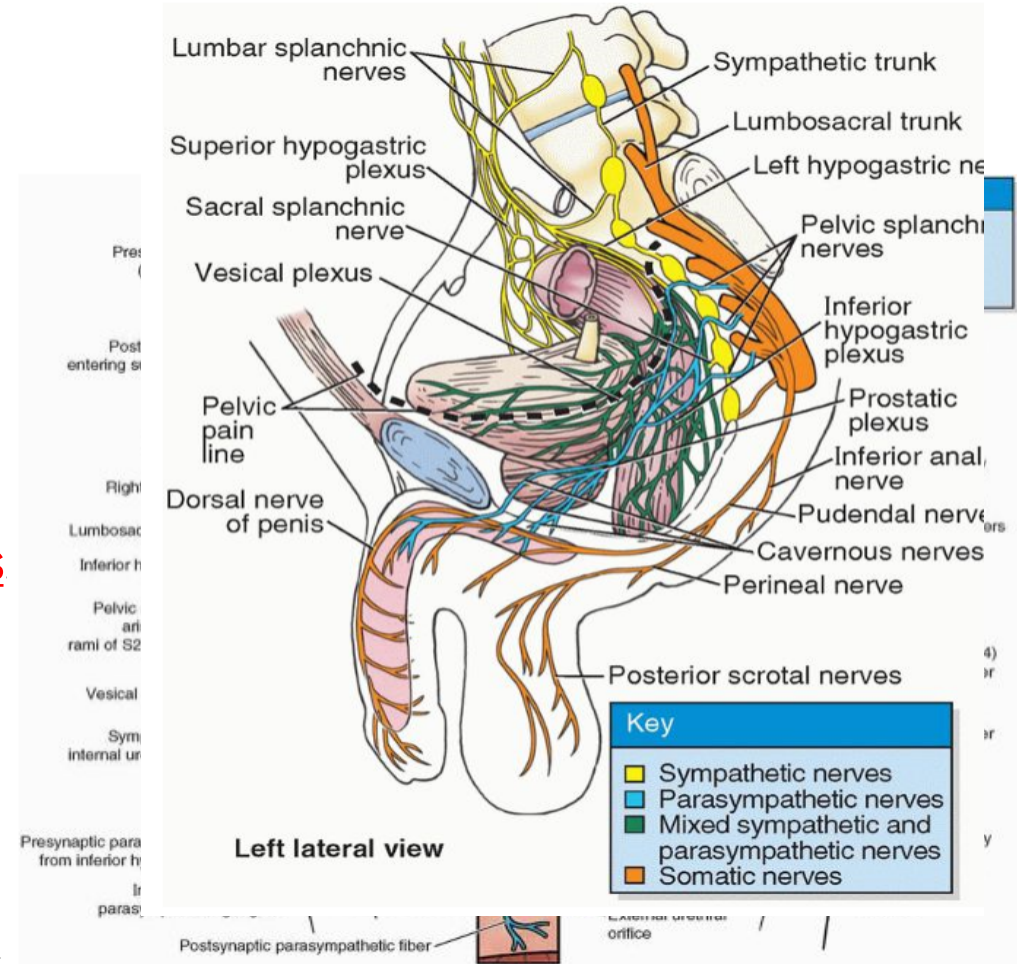
□ Innervation:

• Proximal urethra:

- ✓ Innervation derived from prostatic plexus (mixed sympathetic, parasympathetic, visceral afferent fibers).
- ✓ It arises from inferior hypogastric plexus.

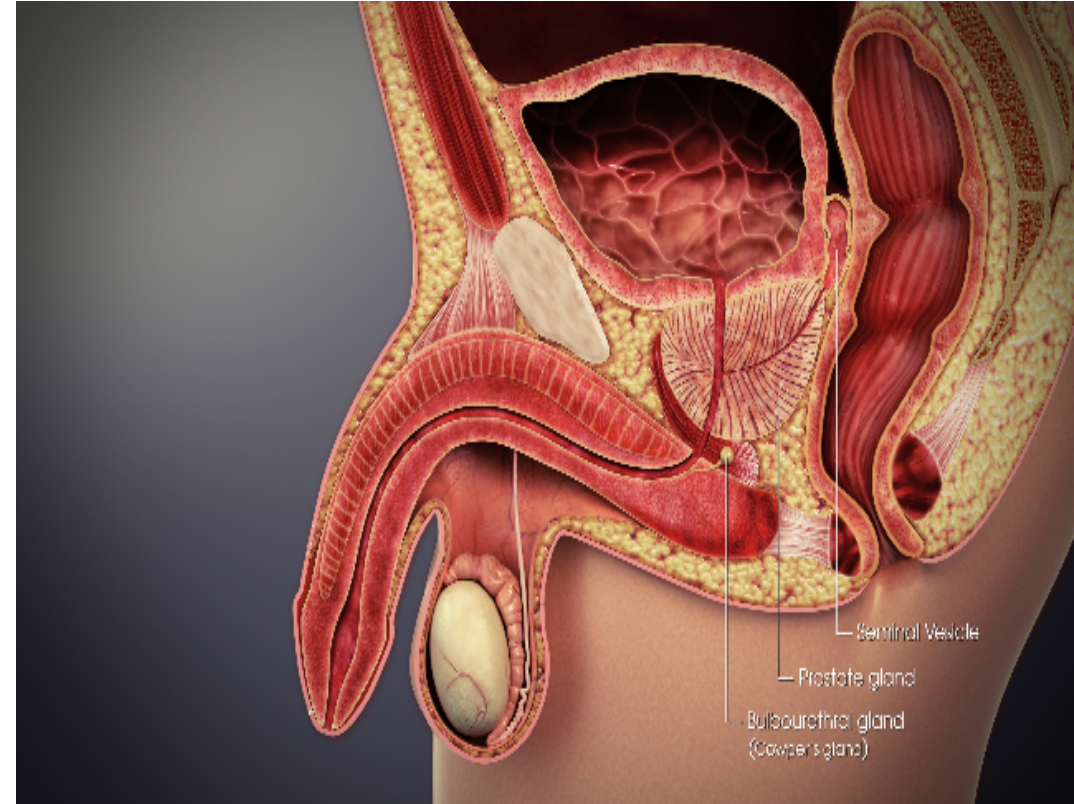
• Distal urethra:

- ✓ **Autonomic** (efferent) innervation via the prostatic nerve plexus.
- ❖ **Sympathetic** innervation is from lumbar part of SC via the lumbar splanchnic nerves
- ❖ **Parasympathetic** innervation is from the sacral level of pelvic splanchnic nerves.
- ✓ **Visceral** (afferent) fibers follow the parasympathetic.
- Dorsal nerve of the penis:
 - Branch of the pudendal nerve.
 - Provides somatic innervation of the spongy part of the urethra.



BULBOURETHRAL GLANDS

- ✓ Also called Cowper glands.
- ✓ Two pea-size glands.
- ✓ Location:
 - ❑ Lie posterolateral to intermediate part of urethra.
 - ❑ Embedded within the external urethral sphincter.
- ✓ Their ducts pass through the perineal membrane.
- ✓ With intermediate urethra.
- ✓ They open into the proximal part of the spongy urethra in the bulb of the penis.
- ✓ Their mucus-like secretion enters the urethra during sexual arousal.



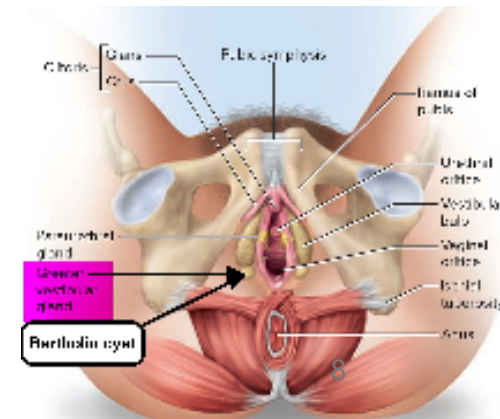
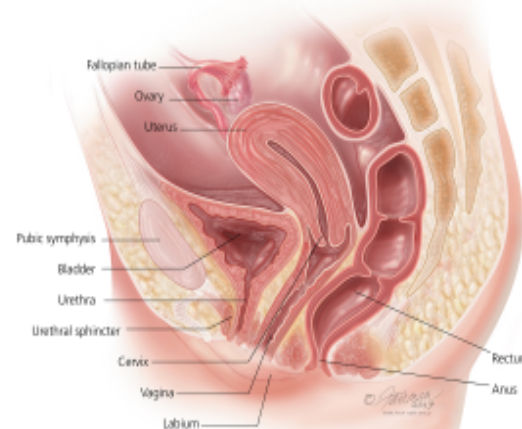
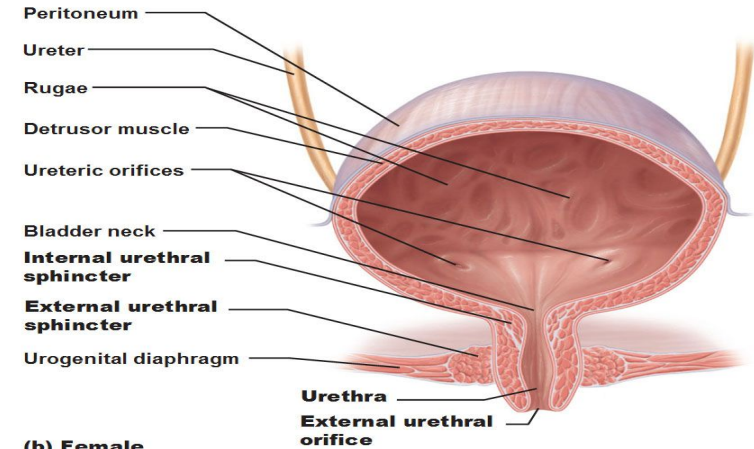
FEMALE URRETHRA

- ❑ Narrow, short tube.
- ❑ About 4 cm long and 6 mm in diameter.
- ❑ It passes anterior- inferiorly from internal urethral orifice of bladder.
- ❑ It runs posterior and then inferior to the pubic symphysis.
- ❑ Ends at external urethral orifice.
- ❑ External urethral orifice is in the vestibule.
- ❑ Directly anterior to the vaginal orifice.

✓ Paraurethral glands:

- o Present in superior part of urethra.
- o One on each side of urethra.
- o They homologs to the prostate.
- o These glands have a common paraurethral duct.
- o Duct opens near the external urethral orifice.

Urinary Bladder and Urethra – Female



THANKS

